

- A list of self-signed (X.509) certificates to constitute the root of trust.
- A list of X.509 certificates to represent the intermediate CAs (certificate authorities) that this provider considers for certificate validation. These certificates ought to be certified by exactly one of the certificates in the root of trust; intermediate CAs are optional parameters.
- A list of X.509 certificates with a verifiable certificate path to exactly one of the certificates of the root of trust to represent the administrators of this MSP; owners of these certificates are authorised to request changes to this MSP configuration (e.g., root CAs and intermediate CAs).
- A list of organisational units that valid members of this MSP should include in their X.509 certificate; this is an optional configuration parameter, used when multiple organisations leverage the same root of trust and intermediate CAs, and have reserved an Organisational unit (OU) field for their members.
- A list of certificate revocation lists (CRLs), each corresponding to exactly one of the listed (intermediate or root) MSP certificate authorities; this is an optional parameter.
- A list of self-signed (X.509) certificates to constitute the Transport Layer Security (TLS) root of trust for the TLS certificate.
- A list of X.509 certificates to represent intermediate TLS CAs that this provider considers. These certificates ought to be certified by exactly one of the certificates in the TLS root of trust; intermediate CAs are optional parameters.

Valid identities for this MSP instance are required to satisfy the following conditions:

- They are in the form of X.509 certificates with a verifiable certificate path to exactly one of the root of trust certificates.
- They are not included in any CRL.
- And they list one or more of the organisational units of the MSP configuration in the OU field of their X.509 certificate structure.

In addition to verification related parameters, for the MSP to enable the node on which it is instantiated to sign or authenticate, one needs to specify the following:

- The signing key used for signing by the node (currently only ECDSA keys are supported), and
- The node's X.509 certificate, which is a valid identity under the verification parameters of this MSP.

How to generate MSP certificates and their signing keys

To generate X.509 certificates to feed its MSP configuration, the application can use Openssl. I'd like to emphasise that in Hyperledger Fabric there is no support for certificates including RSA keys. Alternatively, one can use the cryptogen tool, provided by Hyperledger Fabric. The Hyperledger Fabric

CA can also be used to generate the keys and certificates needed to configure an MSP.

Setting up MSP on the peer and orderer side

To set up a local MSP (for either a peer or an orderer), the administrator should create a folder; e.g., `$MY_PATH/mspconfig` that contains the following six sub-folders and a file:

- The folder *admincerts* to include *PEM* files, each corresponding to an administrator certificate.
- The folder *cacerts* to include *PEM* files, each corresponding to a root CA's certificate.
- Optionally, the folder *intermediatecerts* to include *PEM* files, each corresponding to an intermediate CA's certificate.
- This too is optional — the file *config.yaml* to configure the supported organisational units and identity classifications (see the respective sections below).
- Optionally, the folder *crls* to include the considered CRLs.
- The folder *keystore* to include a *PEM* file with the node's signing key; let me emphasise that currently, RSA keys are not supported.
- The folder *signcerts* to include a *PEM* file with the node's X.509 certificate.
- Optionally, the folder *tlscacerts* to include *PEM* files, each corresponding to a TLS root CA's certificate.
- Optionally, a folder *tlsintermediatecerts* to include *PEM* files, each corresponding to an intermediate TLS CA's certificate.

In the configuration file of the node (the *core.yaml* file for the peer and *orderer.yaml* for the orderer), one needs to specify the path to the *mspconfig* folder and the MSP identifier of the node's MSP. The path to the *mspconfig* folder is expected to be relative to *FABRIC_CFG_PATH* and is provided as the value of the parameter *mspConfigPath* for the peer and *LocalMSPDir* for the orderer. The identifier of the node's MSP is provided as a value of the parameter *localMspId* for the peer and *LocalMSPID* for the orderer. These variables can be overridden via the environment using the CORE prefix for the peer, e.g., *CORE_PEER_LOCALMSPID*, and the ORDERER prefix for the orderer, e.g., *ORDERER_GENERAL_LOCALMSPID*.



Note: For the orderer setup, one needs to generate and provide the genesis block of the system channel to the orderer.

END

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